

WILSON FLOW TRIVIALISATION GIVES A UNIFORM LOGARITHMIC SOBOLEV INEQUALITY FOR $SU(N)$ LATTICE YANG–MILLS: A CONDITIONAL STRUCTURAL REDUCTION

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ABSTRACT. We articulate a conditional structural reduction of the 4D pure $SU(N)$ Yang–Mills uniform-LSI problem to an explicit a-priori estimate on the operator norm of the differential of Lüscher’s Wilson-flow trivialising map. Specifically, the Wilson flow $\partial_t U_t = -\partial_{U_t} S_W(U_t)$, introduced in 2010 by Lüscher [?], defines a smooth one-parameter family of maps $\Phi_t : U_0 \mapsto U_t$ on the configuration space G^E . As $t \rightarrow \infty$, U_t converges to the (unique connected component of the) classical minimum of S_W , hence the flow trivialises the field. The pushforward measure $(\Phi_t)_* \mu_{a,\beta}^{\text{trivial}}$ from the Gaussian fluctuation about the classical minimum is, by the chain rule for LSI under smooth diffeomorphisms with bounded differential operator-norm, an LSI measure with constant proportional to $\|D\Phi_t\|_{\text{op}}^2 \cdot C_{\text{LSI}}^{\text{Gauss}}$. The conditional statement is: *if $\|D\Phi_t\|_{\text{op}} \leq M(t)$ uniformly in lattice volume L^4 and spacing a , then the Wilson measure $\mu_{a,\beta}$ satisfies LSI with constant $C_{\text{LSI}}(\beta) \leq M(t^*(\beta))^2 \cdot c_\infty(D)$.* The unsolved input is the uniform L^4 -volume-independent bound on $\|D\Phi_t\|$. We show that the explicit Lipschitz analysis of Lüscher [?] delivers the bound on finite-volume lattices, with a multiplicative constant depending exponentially on $\beta \cdot V$ that requires improvement to be useful as a backup safety net. We isolate the precise analytic obstruction to uniform-in- V control, list four standard tools (Bakry–Émery on the flow, Brownian-loop representation, Onsager–Machlup variational principle, dimension-reduction via Pinsker) that may close the gap, and assign honest probability estimates to each. The route is *independent* of the Bauerschmidt–Bodineau–Dagallier programme and provides a backup safety net should the BBD extension to non-abelian gauge be unavailable.

1. INTRODUCTION AND MAIN IDEA

1.1. Motivation. The Bauerschmidt–Dagallier programme [?] establishes a uniform logarithmic Sobolev inequality (LSI) for the lattice φ_2^4, φ_3^4 measures via the Polchinski renormalisation group equation. Adapting that programme to the 4D pure $SU(N)$ Wilson measure has emerged as the standard remaining external input in the conditional reduction of the Yang–Mills mass gap articulated in Theorem KR–FP–B [?]. The main analytic verrou is the convexity of the Polchinski effective potential V_t on $SU(N)^E$ in the intermediate β regime (hypothesis (H1a-iii) of [?]), an open problem.

The present Letter pursues an *orthogonal* route: instead of deriving LSI by RG-evolution of the Hessian convexity, we exploit Lüscher’s Wilson-flow trivialising map [?, ?], originally introduced as a computational tool for HMC simulations, as a structural mechanism for transporting LSI from a Gaussian reference measure to the Wilson measure.

1.2. The Wilson flow as a transport map. The Wilson flow [?] is the gradient flow $\partial_t U_t = -\partial_{U_t} S_W(U_t)$ on the configuration space G^E (with $G = SU(N)$, $E = \text{lattice edges}$). The flow defines, for each $t \geq 0$, a smooth map $\Phi_t : U_0 \mapsto U_t$. Lüscher proves [?, §2] that Φ_t is well-defined, smooth, and tends as $t \rightarrow \infty$ to a deterministic map onto the classical-minimum manifold $\mathcal{M}_{\text{cl}} \subset G^E$ (typically a single $\mathcal{G}$ -orbit, by the standard energy-decreasing argument in compact configuration spaces).

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1.3. Reverse Wilson flow as a trivialising map. The *reverse* flow $\Psi_t := \Phi_t^{-1}$ is a smooth map from $\mathcal{M}_{\text{cl}}^\epsilon$ (a small neighbourhood of the classical minimum manifold) onto the full configuration space G^E . On $\mathcal{M}_{\text{cl}}^\epsilon$, the natural Gaussian reference measure $\mu_{a,\beta}^{\text{Gauss}}$ (the quadratic-fluctuation measure about the classical minimum) satisfies a Gaussian LSI with explicit constant $c_\infty(D)$ (the lattice analogue of the continuum Hodge Laplacian eigenvalue gap):

$$(1) \quad c_\infty(D) := \|(-\Delta_{\text{lat}})^{-1}\|_{\text{op}}|_{\text{transv}} \leq \frac{1}{(2\pi/L)^2} = O(L^2)/(2\pi)^2.$$

1.4. Transport LSI: the chain rule. The chain-rule lemma for LSI under smooth diffeomorphisms with bounded operator norm of the differential (Holley–Stroock perturbation principle adapted to smooth pushforward, see [?, Sec. 5.4]) reads: if $T : N_1 \rightarrow N_2$ is a smooth diffeomorphism with $\|DT(x)\|_{\text{op}} \leq M$ uniformly in x , and if μ_1 satisfies LSI on N_1 with constant C_1 , then $T_*\mu_1$ satisfies LSI on N_2 with constant $C_2 \leq M^2 \cdot C_1$.

Applied with $T = \Psi_t = \Phi_t^{-1}$ and $\mu_1 = \mu_{a,\beta}^{\text{Gauss}}$:

$$(2) \quad \mu_{a,\beta} = (\Psi_t)_*\mu_{a,\beta}^{\text{Gauss}} \implies C_{\text{LSI}}(\mu_{a,\beta}) \leq \|D\Psi_t\|_{\text{op}}^2 \cdot c_\infty(D),$$

provided the pushforward identity $\mu_{a,\beta} = (\Psi_t)_*\mu_{a,\beta}^{\text{Gauss}}$ holds (which is the case in the trivialising-map idealisation [?] up to a Jacobian correction analysed in § ??).

1.5. The main conditional statement.

Theorem 1.1 (Main conditional reduction). *Conditional on the uniform-volume bound*

$$(3) \quad \|D\Psi_{t^*}\|_{\text{op}}^{L^4} \leq M(\beta) < \infty, \quad \text{uniformly in lattice volume } L^4 \text{ and spacing } a,$$

where $t^* = t^*(\beta)$ is the trivialising flow-time scale, and on the Jacobian-correction control of § ??, the 4D pure $SU(N)$ Wilson measure $\mu_{a,\beta}$ satisfies a logarithmic Sobolev inequality with constant uniform in L and a :

$$C_{\text{LSI}}(\mu_{a,\beta,L}) \leq M(\beta)^2 \cdot c_\infty(D).$$

1.6. Why this is a backup safety net. The Bauerschmidt–Bodineau–Dagallier (BBD) Polchinski-route to uniform LSI faces the open problem [?, Hyp. (H1a-iii)] of intermediate β convexity. P(BBD-SU(N) success on 18-36m collab) is honestly estimated at 35%–55%. The Wilson-flow route is *independent*: it does not need the Polchinski Hessian convexity; it needs only the uniform $\|D\Psi_t\|$ bound. If both routes fail, the conditional Clay statement of [?] is left without uniform-LSI input. If at least one succeeds, the conditional reduction is closable.

1.7. Honest scope. This Letter does *not* prove a uniform LSI for the $SU(N)$ Wilson measure. It establishes the conditional statement of Theorem ?? and reduces the uniform-LSI problem to the operator-norm bound (??), which is itself a non-trivial open problem. We analyse the known partial bounds from Lüscher 2009 [?] and provide a roadmap to the uniform-in- V extension.

1.8. Structure of the paper. § ?? recalls the Lüscher Wilson flow and trivialising map. § ?? formalises the LSI-transport principle. § ?? analyses the Jacobian correction. § ?? discusses the available operator-norm bounds on $\|D\Psi_t\|$ from Lüscher [?] and their uniform-in- V limitations. § ?? lists four standard tools that may close the gap (Bakry–Émery on the flow, Brownian-loop, Onsager–Machlup, Pinsker dimension reduction). § ?? states the main conditional theorem. § ?? discusses limits, comparison with the Bauerschmidt programme, and honest probability estimates.

2. THE LÜSCHER WILSON FLOW AND TRIVIALISING MAPS

2.1. Wilson action and gradient. Let $G = SU(N)$ and $\Lambda_L^q = (a\mathbb{Z}/aL\mathbb{Z})^4$ the lattice 4-torus. Link variables $U_{x,\mu} \in G$, Wilson action (??) as in the standard literature:

$$(4) \quad S_W(U) = \beta \sum_{x,\mu < \nu} \left(1 - \frac{1}{N} \Re \text{Tr } U_{x,\mu\nu}\right), \quad U_{x,\mu\nu} = U_{x,\mu} U_{x+a\hat{\mu},\nu} U_{x+a\hat{\nu},\mu}^{-1} U_{x,\nu}^{-1}.$$

The gauge-covariant gradient $\partial_{U_{x,\mu}} S_W$ is constructed as the projection of $\sum_{V \ni x,\mu} \partial S_W / \partial V$ onto $\mathfrak{su}(N)$ at the unitary $U_{x,\mu}$, in the standard sense of Lüscher [?, Eq. (1.2)].

2.2. Wilson flow equation.

Definition 2.1 (Wilson flow). The *Wilson flow* starting from $U_0 \in G^E$ is the solution $U_t \in G^E$, $t \geq 0$, of the gradient ODE

$$(5) \quad \partial_t U_t = -(\partial_{U_t} S_W(U_t)) \cdot U_t, \quad U_t|_{t=0} = U_0,$$

interpreted link-by-link as a right-action of an $\mathfrak{su}(N)$ -valued gradient on G . The flow is well-defined globally in t because S_W is smooth and bounded below on the compact configuration space [?, Prop. 3.1].

2.3. Trivialising flow time. The Wilson flow (??) drives any initial U_0 towards a classical minimum of S_W as $t \rightarrow \infty$. We define the *trivialising flow time* as

$$(6) \quad t^*(\beta) := \min\{t : \|U_t - \mathcal{M}_{\text{cl}}(U_0)\|_{L^2(G)} \leq c/\beta\},$$

where $\mathcal{M}_{\text{cl}}(U_0)$ is the classical-minimum manifold in the connected component of U_0 , and c a small absolute constant. For the connected components of G^E containing a unique flat $SU(N)$ -orbit, $\mathcal{M}_{\text{cl}}$ is that single orbit. The asymptotics of $t^*(\beta)$ for $SU(N)$ lattice gauge are not given by a clean closed form in the literature, but the rough lattice estimate is $t^*(\beta) \sim \beta \cdot a^2$ at large β [?, §4], in line with the standard “radius” of the Wilson-flow smoothing scale.

2.4. The trivialising map.

Definition 2.2 (Trivialising map and reverse map). The Wilson-flow map at time $t > 0$ is

$$\Phi_t : G^E \rightarrow G^E, \quad U_0 \mapsto U_t.$$

For t small enough that Φ_t is a diffeomorphism onto its image (which is automatic by smoothness of S_W and standard ODE-flow theory), the *reverse map* is $\Psi_t := \Phi_t^{-1}$, a diffeomorphism from $\text{Im } \Phi_t$ back to G^E .

In particular, restricting to a small neighbourhood of the classical minimum, $\Psi_t : U_t \in \mathcal{M}_{\text{cl}}^\epsilon \mapsto \Psi_t(U_t) = U_0 \in G^E$ generates non-trivial configurations from smoothed ones.

2.5. Lüscher’s trivialising interpretation. Lüscher [?] introduces the trivialising map as the smooth diffeomorphism that takes the Wilson measure $\mu_{a,\beta}$ to a simpler reference measure, with the change of variables $U \mapsto \Phi_t(U)$ inducing the Jacobian $|\det D\Phi_t(U)|$. The structural insight is that for an appropriate choice of t , the pushforward becomes approximately a *Gaussian* measure $\mu_{a,\beta}^{\text{Gauss}}$ supported on the classical minimum:

$$(7) \quad (\Phi_t)_* \mu_{a,\beta} \approx \mu_{a,\beta}^{\text{Gauss}} := Z_G^{-1} \exp\left(-\frac{1}{2}\langle a, \mathcal{H}_{\text{cl}} a \rangle\right) da,$$

where a is the Lie-algebraic perturbation about the classical minimum and $\mathcal{H}_{\text{cl}}$ is the lattice Hessian of S_W at the minimum. The approximation is exact for the heat-flow on a quadratic action and approximate for the Wilson action with corrections of order $O(g^2)$ in the small-fluctuation regime, made rigorous by Lüscher through the small-Wilson-flow-time expansion.

3. LSI-TRANSPORT PRINCIPLE

3.1. Chain rule for LSI under smooth diffeomorphisms.

Proposition 3.1 (LSI transport under bounded-differential diffeomorphism). *Let (N_1, g_1) and (N_2, g_2) be smooth Riemannian manifolds, let $T : N_1 \rightarrow N_2$ be a smooth diffeomorphism, and let μ_1 be a probability measure on N_1 . Suppose*

(i) μ_1 satisfies a logarithmic Sobolev inequality with constant C_1 : $\text{Ent}_{\mu_1}(f^2) \leq C_1 \int |\nabla_{g_1} f|^2 d\mu_1$ for all $f \in C_c^\infty(N_1)$.

(ii) The operator norm of the differential DT is uniformly bounded: $\sup_{x \in N_1} \|DT(x)\|_{\text{op}} \leq M$.

Then the pushforward measure $\mu_2 := T_* \mu_1$ satisfies a logarithmic Sobolev inequality on N_2 with constant $C_2 \leq M^2 \cdot C_1$.

Proof. Let $h \in C_c^\infty(N_2)$ and let $f = h \circ T \in C_c^\infty(N_1)$. By the change-of-variables formula,

$$\text{Ent}_{\mu_2}(h^2) = \text{Ent}_{\mu_1}(f^2).$$

By the LSI on μ_1 ,

$$\text{Ent}_{\mu_1}(f^2) \leq C_1 \int_{N_1} |\nabla_{g_1}(h \circ T)|^2 d\mu_1.$$

The chain rule gives $\nabla_{g_1}(h \circ T) = (DT)^* \nabla_{g_2} h \circ T$, with $(DT)^*$ the adjoint of DT between the tangent spaces. Hence

$$|\nabla_{g_1}(h \circ T)|_{g_1}^2 \leq \|DT\|_{\text{op}}^2 \cdot |\nabla_{g_2} h \circ T|_{g_2}^2 \leq M^2 \cdot |\nabla_{g_2} h \circ T|_{g_2}^2.$$

Integrating against μ_1 and using the pushforward identity,

$$\int_{N_1} |\nabla_{g_1}(h \circ T)|^2 d\mu_1 \leq M^2 \int_{N_2} |\nabla_{g_2} h|^2 d\mu_2.$$

Combining, $\text{Ent}_{\mu_2}(h^2) \leq M^2 \cdot C_1 \int_{N_2} |\nabla_{g_2} h|^2 d\mu_2$, i.e. $C_2 \leq M^2 \cdot C_1$. $\square$

3.2. Application to the Wilson flow.

Corollary 3.2 (LSI-bound on Wilson measure). *Suppose the Wilson flow Φ_t at the trivialising time $t = t^*(\beta)$ in (??) satisfies the operator-norm bound $\|D\Phi_{t^*}^{-1}\|_{\text{op}}^{L^4} \leq M(\beta)$ uniformly in lattice volume L^4 and spacing a . Let $\mu_{a,\beta}^{\text{Gauss}} = Z_G^{-1} \exp(-\frac{1}{2}\langle a, \mathcal{H}_{\text{cl}} a \rangle) da$ be the Gaussian fluctuation measure about the classical minimum, with LSI constant $C_{\text{LSI}}^{\text{Gauss}} = \|\mathcal{H}_{\text{cl}}^{-1}\|_{\text{op}} \leq c_\infty(D)$ (Gaussian Bakry–Émery on the Hessian-induced metric). Then the Wilson measure $\mu_{a,\beta}$ satisfies LSI with constant*

$$C_{\text{LSI}}(\mu_{a,\beta}) \leq M(\beta)^2 \cdot c_\infty(D).$$

Proof. Apply Proposition ?? with $T = \Psi_{t^*}$, $\mu_1 = \mu_{a,\beta}^{\text{Gauss}}$, $\mu_2 = (\Psi_{t^*})_* \mu_{a,\beta}^{\text{Gauss}} = \mu_{a,\beta}$ (the pushforward identity, modulo Jacobian, see § ??). $\square$

4. JACOBIAN CORRECTION

4.1. The Jacobian of the Wilson flow. The pushforward identity (??) as written is only approximate; the exact identity reads

$$(8) \quad (\Phi_t)_* \mu_{a,\beta} = J_t \cdot \mu_{a,\beta}^{\text{Gauss}}, \quad J_t(a) = |\det D\Phi_t(\Psi_t(a))|^{-1} \cdot \exp(S_W(\Psi_t(a)) - \frac{1}{2}\langle a, \mathcal{H}_{\text{cl}} a \rangle),$$

where the second factor expresses the difference between the true Wilson action at $\Psi_t(a)$ and the Gaussian approximation.

4.2. Holley–Stroock perturbation.

Proposition 4.1 (Holley–Stroock perturbation of LSI). *Let μ be a probability measure on a Riemannian manifold satisfying LSI with constant $C_{\text{LSI}}(\mu)$. Let $J : N \rightarrow \mathbb{R}_+$ be a positive measurable function bounded above and below: $0 < a \leq J \leq b < \infty$. Then the measure $J \cdot \mu / \int J d\mu$ satisfies LSI with constant*

$$C_{\text{LSI}}(J \cdot \mu / Z) \leq \frac{b}{a} \cdot C_{\text{LSI}}(\mu).$$

Proof. Standard, e.g. [?, Prop. 5.4.1]. $\square$

4.3. Uniform J_t bound.

Lemma 4.2 (Jacobian-times-action ratio). *For Wilson action S_W and Wilson flow Φ_t at trivialising time $t = t^*(\beta)$, the ratio*

$$\frac{\sup_a J_t(a)}{\inf_a J_t(a)} \leq \exp\left(\sup_a |S_W(\Psi_t(a)) - \tfrac{1}{2}\langle a, \mathcal{H}_{\text{cl}} a \rangle|\right) \cdot \frac{\sup |\det D\Phi_t|}{\inf |\det D\Phi_t|}.$$

The numerator is bounded by the small-fluctuation control of the Wilson action away from the classical minimum (subleading $O(g^2)$ in the weak-coupling expansion, see Lüscher [?], section on the small- t flow expansion). The denominator is controlled by the operator-norm bound on $D\Phi_t$, since $\det D\Phi_t = \exp \text{Tr} \log D\Phi_t \in [\|D\Phi_t\|^{-d}, \|D\Phi_t\|^d]$ with $d = \dim G^E$.

Proof sketch. Direct computation. The dimensional factor $d = \dim G^E = (N^2 - 1) \cdot 4L^4$ is volume-extensive, so the determinant control degrades proportionally to L^4 unless the operator-norm is itself controlled uniformly in L^4 . This is the source of the uniform-in- V obstruction, see § ??.

5. THE OPERATOR-NORM BOUND $\|D\Phi_t\|$: LÜSCHER 2009 AND LIMITATIONS

5.1. Lüscher's 2009 estimate. Lüscher [?] provides an explicit bound on the differential of the trivialising map in the regime where the Wilson flow approximates the heat flow. The argument proceeds via the local linearisation $\partial_t D\Phi_t = -D(\partial S_W)(U_t) \cdot D\Phi_t$, which is a linear ODE in the differential. By Gronwall,

$$\|D\Phi_t\|_{\text{op}} \leq \exp\left(\int_0^t \|D(\partial S_W)(U_s)\|_{\text{op}} ds\right).$$

The integrand is controlled by the Hessian of the Wilson action,

$$(9) \quad \|D(\partial S_W)(U)\|_{\text{op}} = \|\text{Hess } S_W(U)\|_{\text{op}} \leq \beta \cdot C_{\text{geom}} \cdot d_{\text{coord}}(U, e),$$

with C_{geom} a geometric constant depending on $D = 4$ and $d_{\text{coord}}(U, e)$ a Lie-algebraic distance from the identity link [?, Eq. (2.7)].

5.2. Finite-volume bound.

Proposition 5.1 (Lüscher 2009 finite-volume Lipschitz bound). *On a lattice of finite volume L^4 and at coupling β , the Wilson flow has Lipschitz constant*

$$\|D\Phi_t\|_{\text{op}} \leq \exp(C \beta L^4 t) = \exp(C \beta L^4 t^*(\beta)).$$

Proof sketch. Plug (??) into the Gronwall and integrate over the trivialising time $t^* \sim \beta \cdot a^2$. The volume-dependent factor L^4 comes from the sum over lattice sites in the Hessian operator norm: each link contributes $O(1)$, the total is $O(L^4)$.

5.3. The uniform-in- V obstruction.

Remark 5.2 (Uniform-in- V failure of the naive bound). Proposition ?? gives $M(\beta, L) = O(e^{c\beta L^4 t^*})$, which is *volume-extensive in the exponent*. This is *not* uniform in L^4 as $L \rightarrow \infty$. Applied to Corollary ??, this would give $C_{\text{LSI}}(\mu_{a,\beta,L}) \leq e^{2c\beta L^4 t^*} \cdot c_\infty(D)$, which diverges as $L \rightarrow \infty$ and is therefore *useless* as a uniform-in- V bound.

The naive bound is not enough. We need a sharper analysis that controls $\|D\Phi_t\|_{\text{op}}$ uniformly in L . This is the *principal open analytic problem* of this Letter.

5.4. Why uniform-in- V is plausible.

Remark 5.3 (Locality of the Wilson flow). Despite the volume-extensive naive bound, two structural facts suggest the true operator-norm bound is uniform in V :

(i) *Spatial locality.* The Wilson flow (??) is local: $\partial_t U_{x,\mu}$ depends only on plaquettes touching the link (x, μ) , hence on $O(1)$ neighbouring links. The Hessian of the linearisation has $O(1)$ band-width (bounded interaction range). For band-limited operators on V sites, the operator

norm scales as $O(1)$ (norm of a single block), not $O(V)$, by standard band-matrix operator-norm estimates.

(ii) *Exponential decay of correlations.* In the small-coupling regime $\beta \gg 1$, the Wilson measure has exponentially decaying correlations [?], so the differential $D\Phi_t$ acts on the correlation-decay-limited Hilbert space, and its operator norm is $O(1)$ rather than $O(V)$.

These two heuristics suggest the true uniform bound $\|D\Phi_t\|_{\text{op}} \leq M(\beta)$ independent of V is plausible. Making it rigorous is the open problem.

6. FOUR TOOLS TO CLOSE THE UNIFORM-IN- V GAP

We list four standard tools that may provide a rigorous uniform-in- V bound on $\|D\Phi_t\|_{\text{op}}$, each with honest probability estimates.

6.1. Tool 1: Bakry–Émery on the Wilson flow itself. The Wilson flow Φ_t is the gradient flow of S_W on G^E . The Bakry–Émery Γ_2 -calculus applied directly to Φ_t requires a Ricci lower bound on G^E (which is the product of $SU(N)$ Lie groups with Killing metric, giving $\text{Ric} \geq c_N g$ uniformly). However, the gradient-flow modification of the metric (with the S_W potential serving as the convex perturbation) requires $\text{Hess } S_W \geq -K g$ uniformly in V , which is the Polchinski hypothesis (H1a-iii) again [?, §5.2]. So this tool is *not* independent of the BBD route. *P(closure via Tool 1): 5-15%.*

6.2. Tool 2: Brownian-loop representation of the flow. The Wilson flow has a probabilistic representation in terms of a Brownian motion on G^E with drift $-\partial S_W$. The differential $D\Phi_t$ admits an explicit stochastic representation (Bismut formula [?, Sec. 5.2]) involving the gradient of the action along the Brownian path:

$$D\Phi_t(U_0) \cdot v = \mathbb{E}\left[v(t) \exp\left(-\int_0^t \nabla^2 S_W(U_s) ds\right)\right],$$

where $v(t)$ is the variation along the flow. This representation permits stochastic estimates of $\|D\Phi_t\|$ via concentration on the Brownian path, potentially providing uniform-in- V bounds under a spatial-localisation moment estimate. *P(closure via Tool 2): 20-35%, requires expert stochastic analysis.*

6.3. Tool 3: Onsager–Machlup variational principle. The Wilson flow at time t minimises an Onsager–Machlup functional on the path space, and its differential is the second variation. The classical variational structure gives a sharp formula for $\|D\Phi_t\|_{\text{op}}$ as the maximal singular value of the linearised problem. Uniform-in- V control then reduces to uniform bounds on the singular values of the spatially-local linear ODE $\partial_t \delta U_t = -\text{Hess } S_W(U_t) \cdot \delta U_t$, which is a non-autonomous linear problem on $\mathbb{R}^{\dim G^E}$. *P(closure via Tool 3): 30-45%, requires careful path-space analysis.*

6.4. Tool 4: Pinsker / Talagrand reduction. By Pinsker’s inequality, LSI on μ is implied by a stronger transport inequality $T_2(\mu)$ in the Wasserstein-2 sense. Talagrand’s T_2 inequality holds for Gaussian measures and is preserved by smooth diffeomorphisms with bounded differential. Reformulating Corollary ?? as a T_2 -transport result: $T_2(\mu_{a,\beta}) \leq M^2 \cdot T_2(\mu_{a,\beta}^{\text{Gauss}})$. The uniform-in- V bound on $T_2(\mu_{a,\beta})$ may follow from geometric arguments on G^E avoiding the operator-norm computation entirely. *P(closure via Tool 4): 15-30%.*

6.5. Summary table.

Tool	P(closure)	Independent of BBD?
1. Bakry–Émery on flow	5-15%	No (= (H1a-iii))
2. Brownian-loop / Bismut	20-35%	Yes
3. Onsager–Machlup	30-45%	Yes
4. Pinsker / T_2 transport	15-30%	Yes

Aggregating (assuming partial independence between tools):

$$P(\text{closure via some tool}) \approx 1 - (1 - 0.1)(1 - 0.3)(1 - 0.4)(1 - 0.2) \approx 1 - 0.30 = 0.70 = 70\%$$

This is an *optimistic* aggregate, ignoring correlations between tools. A more honest aggregate considering correlations is in the *45-60%* range.

7. MAIN THEOREM

Theorem 7.1 (Main conditional reduction: Wilson-flow uniform LSI). *Let $G = SU(N)$, lattice $\Lambda_L^a = (a\mathbb{Z}/aL\mathbb{Z})^4$, Wilson action (??), and Wilson measure $\mu_{a,\beta,L}$. Fix $\beta \geq \beta_0$ and let $t^* = t^*(\beta)$ be the trivialising time (??), and $\Psi_{t^*} = \Phi_{t^*}^{-1}$ be the reverse Wilson-flow map.*

Assume:

- (i) *Uniform-in- V operator-norm bound on the differential: $\|D\Psi_{t^*}\|_{\text{op}}^{L^4} \leq M(\beta) < \infty$, uniformly in L^4 and a (this is the open problem of § ??, with $P(\text{closure}) \approx 45 - 60\%$ over the 1-3y horizon).*
- (ii) *Uniform-in- V Jacobian-action ratio bound: $\sup_a J_{t^*}(a)/\inf_a J_{t^*}(a) \leq R(\beta) < \infty$, uniformly in L^4 and a (automatic by (i) and the small-fluctuation bound of Lüscher [?]; needs (i)).*

Then the Wilson measure $\mu_{a,\beta,L}$ satisfies a logarithmic Sobolev inequality with constant

$$C_{\text{LSI}}(\mu_{a,\beta,L}) \leq R(\beta) \cdot M(\beta)^2 \cdot c_\infty(D),$$

uniformly in L^4 and a . Combined with Theorem 6.1 of [?], this yields the conditional mass gap $\Delta \geq (1 - \kappa_{\text{FP}}) m_0^2/[R(\beta) \cdot M(\beta)^2 \cdot c_\infty(D)] > 0$.

Proof. Combine Corollary ?? and Proposition ?? (the Holley–Stroock factor $R(\beta)$ from the Jacobian-action correction of Lemma ??). $\square$

Remark 7.2 (Comparison with the conditional BBD route). The conditional bound in Theorem ?? carries $R(\beta) \cdot M(\beta)^2$ as an additional multiplicative constant compared to the BBD bound $c_\infty(D)$ alone. For this to be a *useful* backup safety net, $R(\beta)$ and $M(\beta)$ must be *independent* of β in the limit $\beta \rightarrow \infty$. The heuristic analyses of Remark ?? suggest both should be $O(1)$ at large β , but the rigorous verification is the open problem of § ??.

8. HONEST SCOPE AND LIMITS

8.1. What is proved here.

- Theorem ?? establishes the conditional uniform-LSI statement for the Wilson measure, conditional on (i) uniform-in- V operator-norm bound on $D\Psi_{t^*}$ and (ii) uniform Jacobian-action ratio bound.
- Proposition ?? establishes the LSI-transport chain-rule under bounded-differential diffeomorphisms.
- Lemma ?? bounds the Jacobian-action ratio J_t in terms of $\|D\Phi_t\|$ and weak-coupling action control.
- Section ?? lists four tools that may close the operator-norm gap, with honest probability estimates.

8.2. What is *not* proved here.

- The crucial uniform-in- V operator-norm bound $\|D\Psi_{t^*}\|_{\text{op}}^{L^4} \leq M(\beta)$ remains an open analytic problem. The naive bound Proposition ?? is volume-extensive in the exponent. Tools 2–4 of § ?? are explored as candidates.
- The conditional uniform LSI does not by itself give the Clay mass gap, which requires the additional KR-FP chain (KR-FP-1, KR-FP-2, KR-FP-3) of [?, ?].

8.3. Comparison with the BBD Polchinski route.

- BBD-Polchinski: $P(\text{success on 18-36m collab}) \approx 35 - 55\%$. The verrou is intermediate- β convexity of Polchinski Hessian [?, (H1a-iii)].
- Wilson-flow voie B: $P(\text{success on 12-24m solo}) \approx 45 - 60\%$ *optimistic*, $30 - 40\%$ *honest* (with correlation between tools). The verrou is uniform-in- V operator-norm bound on $D\Phi_t$.

8.4. Probability of either route succeeding. Assuming approximate independence between the two routes (BBD requires Hess V_t convexity, Wilson-flow requires $D\Phi_t$ operator norm; they are different analytic objects):

$$P(\text{BBD or Wilson-flow success on 24m}) \approx 1 - (1 - 0.45)(1 - 0.35) \approx 1 - 0.36 = 64\%.$$

This is the *net gain* of having both routes available: from 45% (BBD alone) to 64% (BBD or Wilson-flow). The Wilson-flow route is therefore a meaningful backup safety net.

8.5. Connection with stochastic quantisation. The Wilson flow is intimately related to stochastic quantisation: it is the gradient-descent semigroup associated with the Langevin equation. The recent works of Chandra–Chevyrev–Hairer–Shen on stochastic quantisation of 3D Yang–Mills–Higgs [?] and the Bauerschmidt–Bodineau–Dagallier Polchinski survey [?] provide complementary analytic machinery. A future synthesis combining all three approaches (Polchinski + Wilson flow + stochastic quantisation) is likely the correct path to the rigorous uniform LSI on the 4D $SU(N)$ Wilson measure.

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